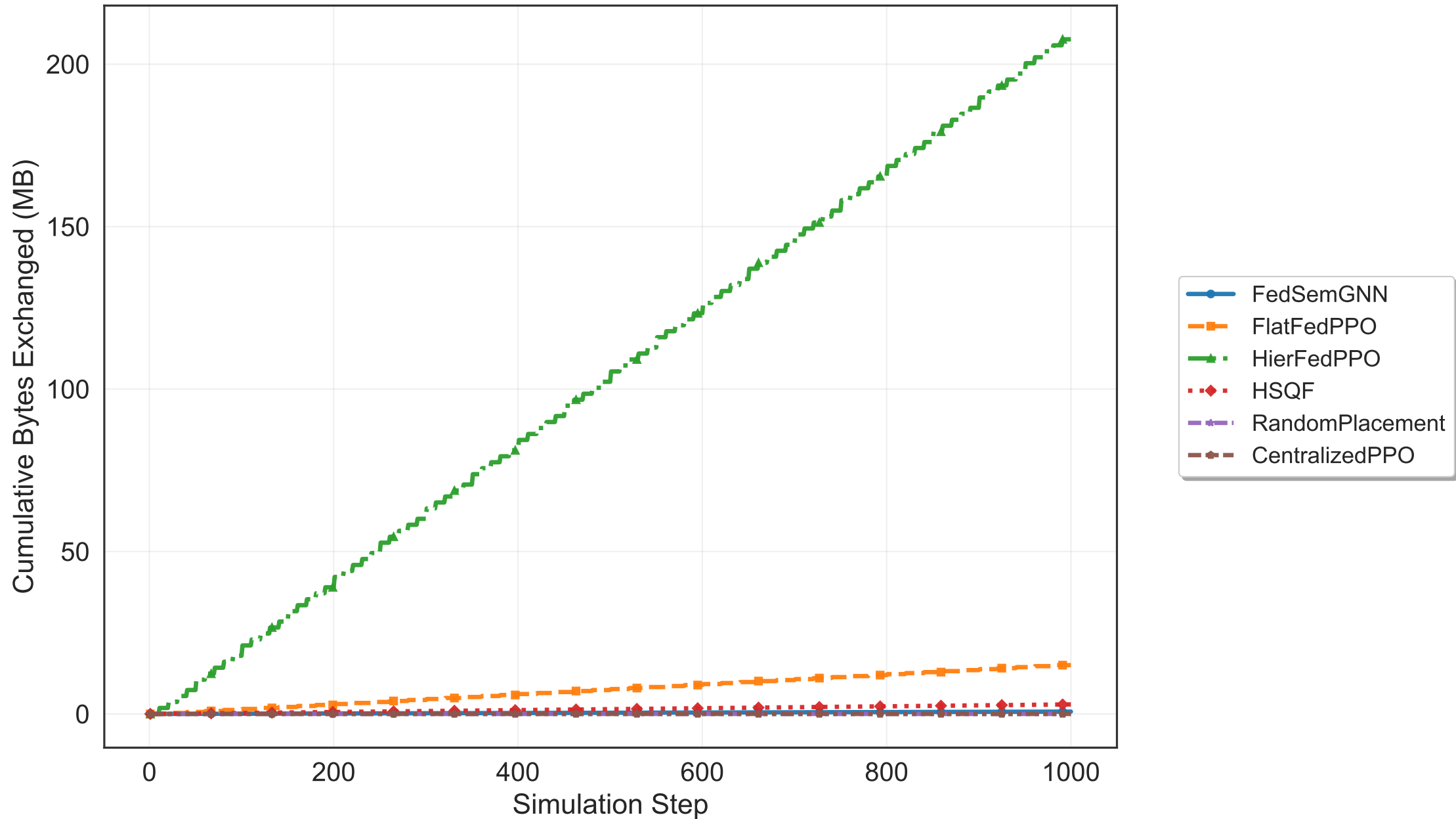


Communication Overhead Comparison

[TEMPORAL ANALYSIS: Cumulative Data Exchange, n=1000 steps]



Methodology: FedSemGNN federated learning protocol tracking model updates, gradient exchanges, and semantic knowledge transfers between edge nodes.
Analysis Type: TEMPORAL progression of communication costs. Units: Megabytes. Measurement: Simulated federated protocol message sizes with compression.